

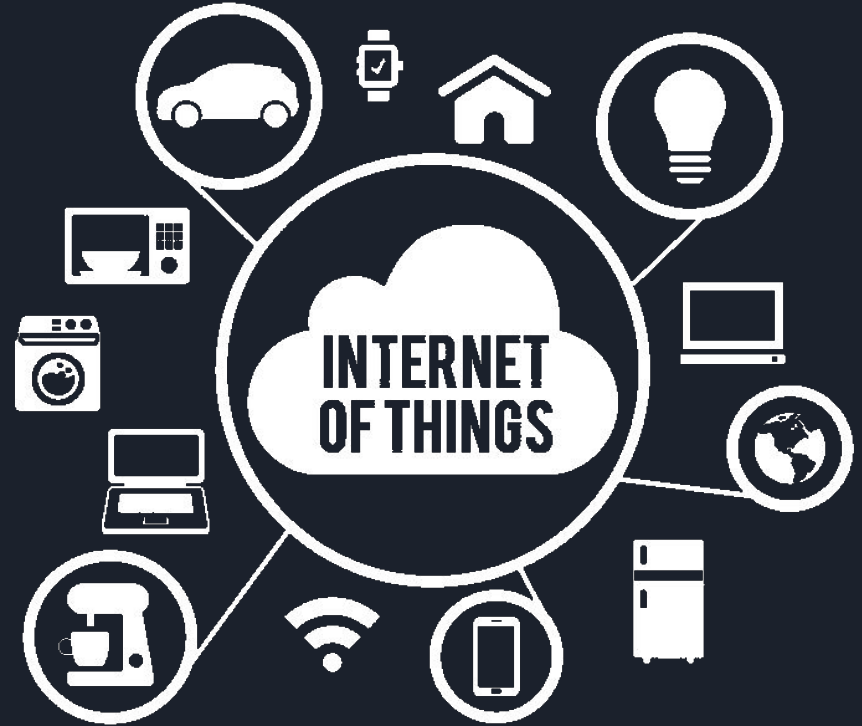


Internet of things & Internet of Everything

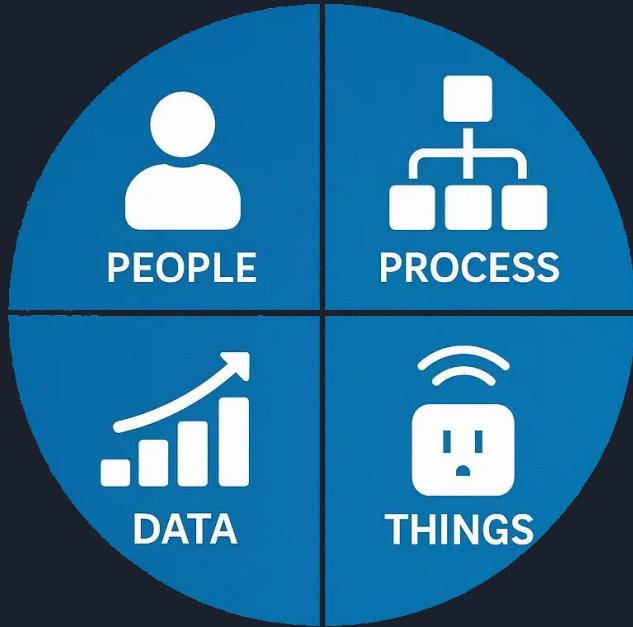
Borys Dąbrowski Kamil Borkowski Jan
Czyżak

What is Internet of Things?

- Network of physical objects connected to the internet
- Devices collect, send, and receive data
- Embedded with sensors, actuators, and connectivity
- Goal: automation, efficiency, insight



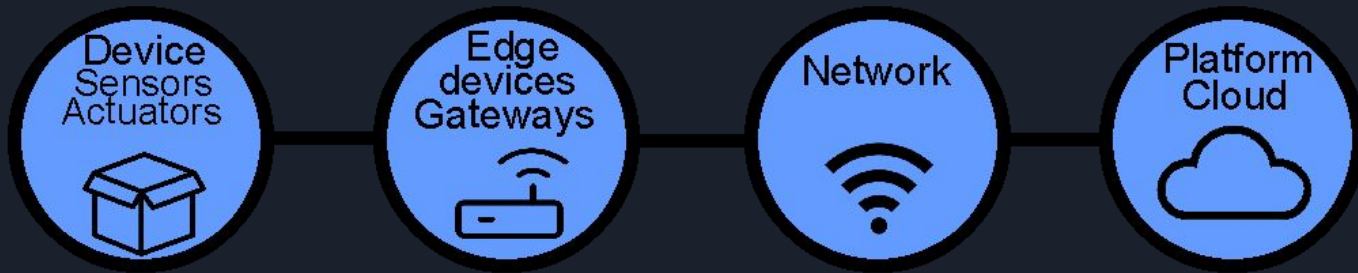
How IoE Differs from IoT



- IoT: focuses on things
- IoE: includes people, processes, data, and things
- IoE = broader concept
- Emphasis on intelligent interactions and contextual decisions

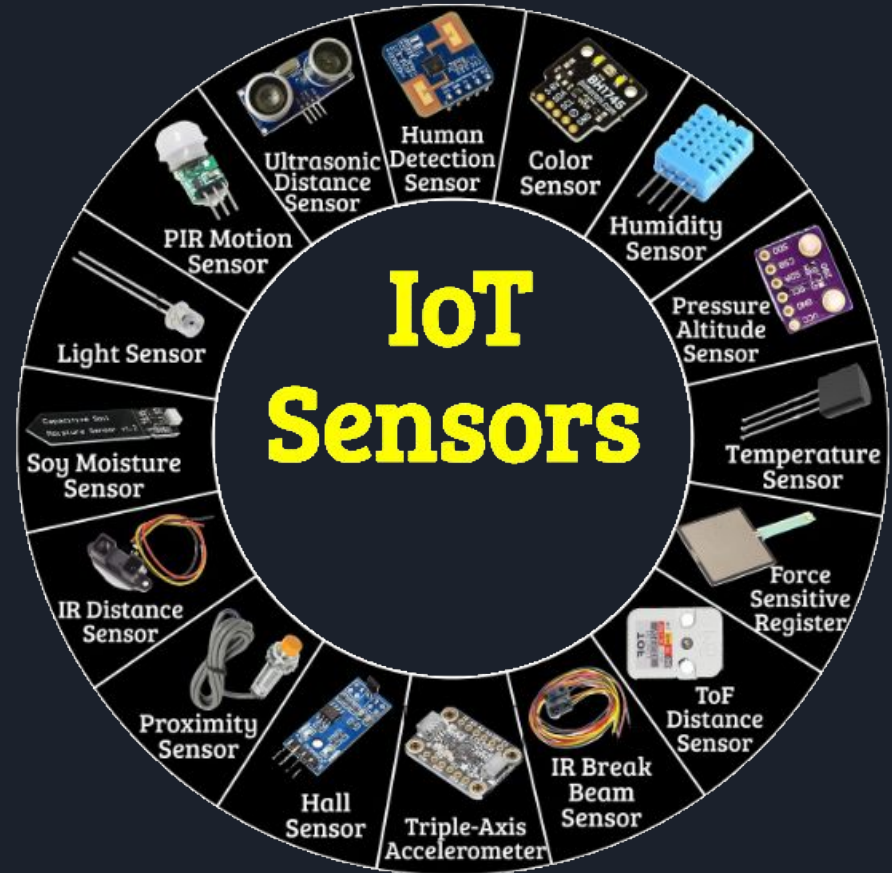
Elements of the IoT Ecosystem

- Devices (sensors & actuators)
- Edge devices & gateways
- Communication networks
- Cloud / IoT platforms



Sensors and Actuators

- Sensors: temperature, humidity, motion, GPS, etc.
- Actuators: motors, switches, valves
- Enable data collection and physical response



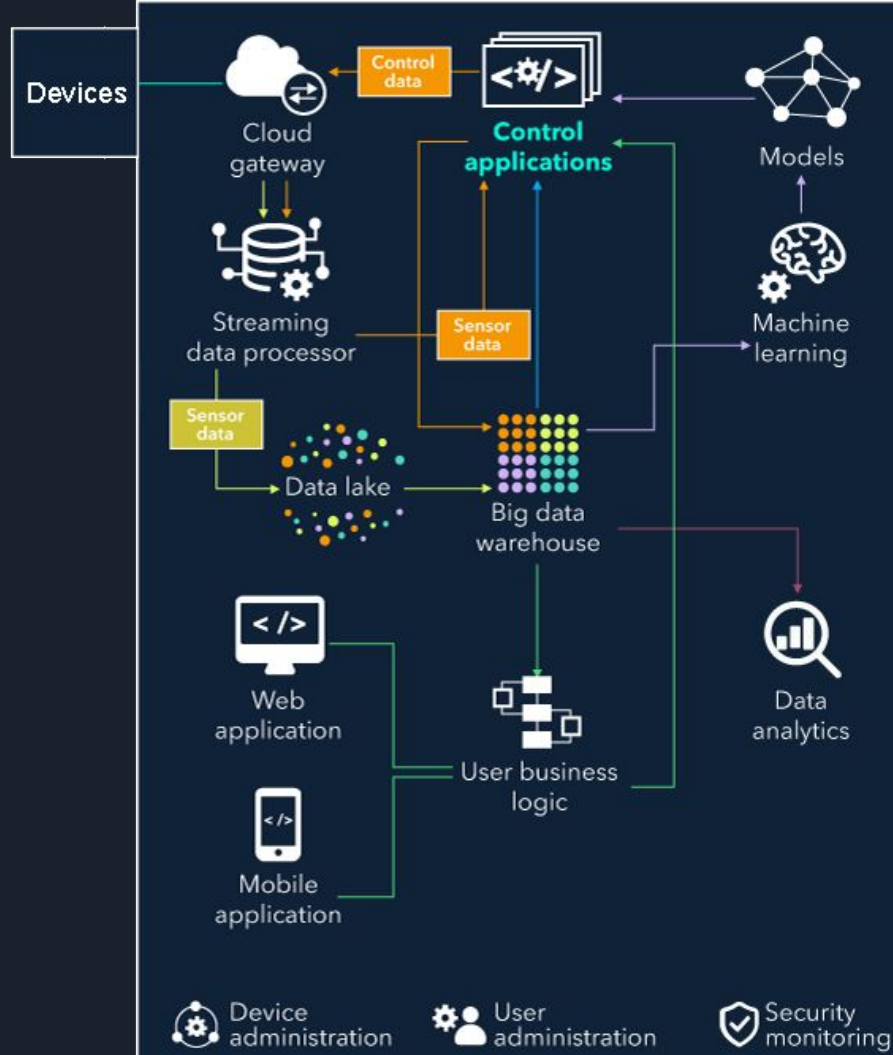


IoT Communication Networks

Technology	Range	Power Consumption	Data Range	Topology	Typical Use
Wi-Fi	20–50 m	Medium–High	High (Mbps)	Star	Smart home devices, cameras
Bluetooth Low Energy (BLE)	5–30 m	Very Low	Low–Medium (Kbps–Mbps)	Star / Mesh	Wearables, beacons, health trackers
Zigbee	10–100 m (mesh extends range)	Low	Low (Kbps)	Mesh	Smart home sensors, lighting, automation
LoRaWAN (Long Range Wide Area Network)	2–15 km	Very Low	Very Low (Kbps)	Star-of-stars	Environmental monitoring, smart agriculture
NB-IoT (Narrowband)	Nation-wide (cellular)	Low	Low (tens to hundreds Kbps)	Star	Smart metering, asset tracking
LTE-M	Nation-wide	Low–Medium	Medium (hundreds Kbps–Mbps)	Star	Mobile IoT, wearables, logistics
5G	100 m–km (depending on band)	Medium	Very High	Star	Autonomous vehicles, real-time applications

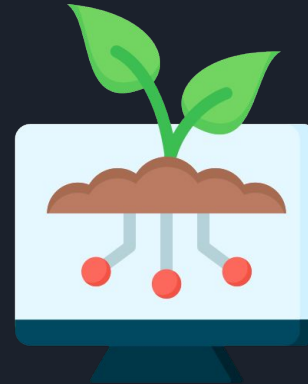
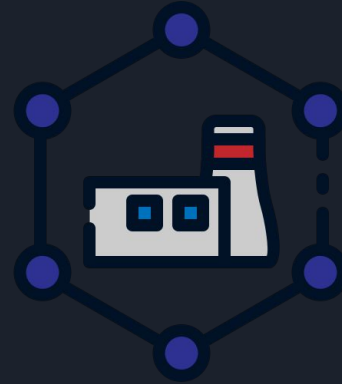
IoT Platforms

- Cloud-based management & analytics
- Data storage & processing
- Device management
- Examples: AWS IoT, Azure IoT Hub, Google IoT Core

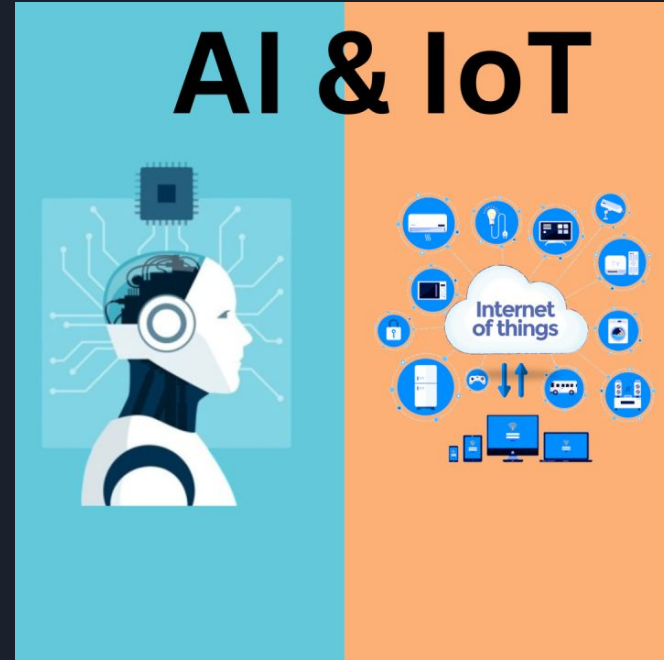


IoT / IoE Use Cases

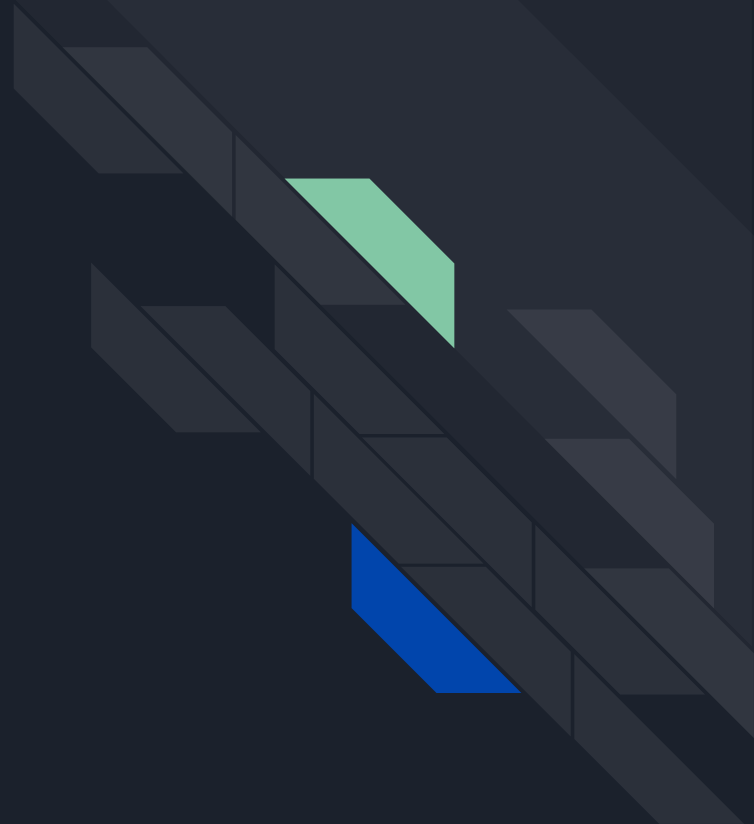
- Smart Homes
- Smart Cities
- Industry (IIoT)
- Healthcare
- Agriculture, Retail, Transportation



- Edge computing
- AI and machine learning integration
- Device miniaturization
- Improved security



Network Infrastructure Requirements

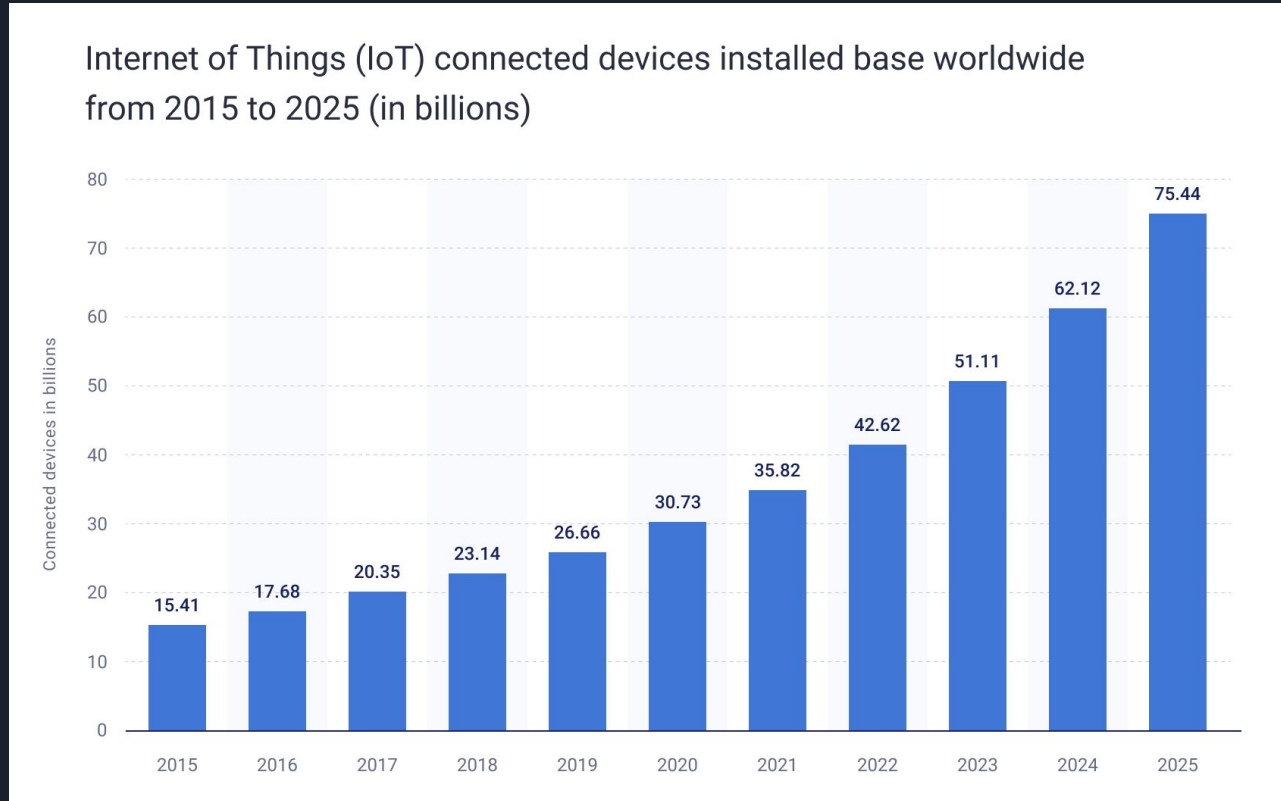


Requirement: network scalability

IoT in industrial use should be able to support increasing number of devices, higher data volumes and more complex traffic patterns with its performance remaining on a constant level.



Growing amount of IoT devices



Demand for bandwidth and low latency

Technology	Range	Data Range
Wi-Fi	20–50 m	High (Mbps)
Bluetooth Low Energy (BLE)	5–30 m	Low–Medium (Kbps–Mbps)
Zigbee	10–100 m (mesh extends range)	Low (Kbps)
LoRaWAN	2–15 km (rural up to 20 km)	Very Low (Kbps)
NB-IoT	Nation-wide (cellular)	Low (tens to hundreds Kbps)
LTE-M	Nation-wide	Medium (hundreds Kbps–Mbps)
5G (mMTC/URLLC)	100 m–km (depending on band)	Very High

Strategies

- IPv6 adoption - address space for billions of devices
- Usage of hierarchical architecture of computing layers
- Efficient data management
- Right choice of communication protocols

IPv6 Address

2001 : 0DC8 : E004 : 0001 : 0000 : 0000 : 0000 : F00A

16 bits : 16 bits : 16 bits : 16 bits : 16 bits : 16 bits : 16 bits : 16 bits

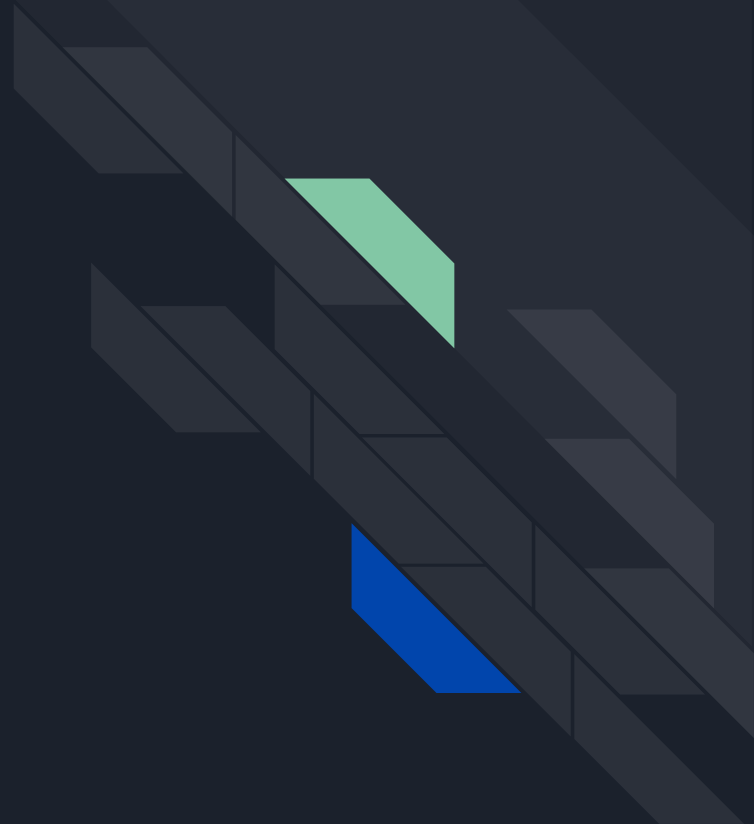
128 bits

IPv4 - ~4.3 billion addresses

IPv6 - 2^{128} possibilities

It matters for scalability!!!

IoT Computing layers



Cloud

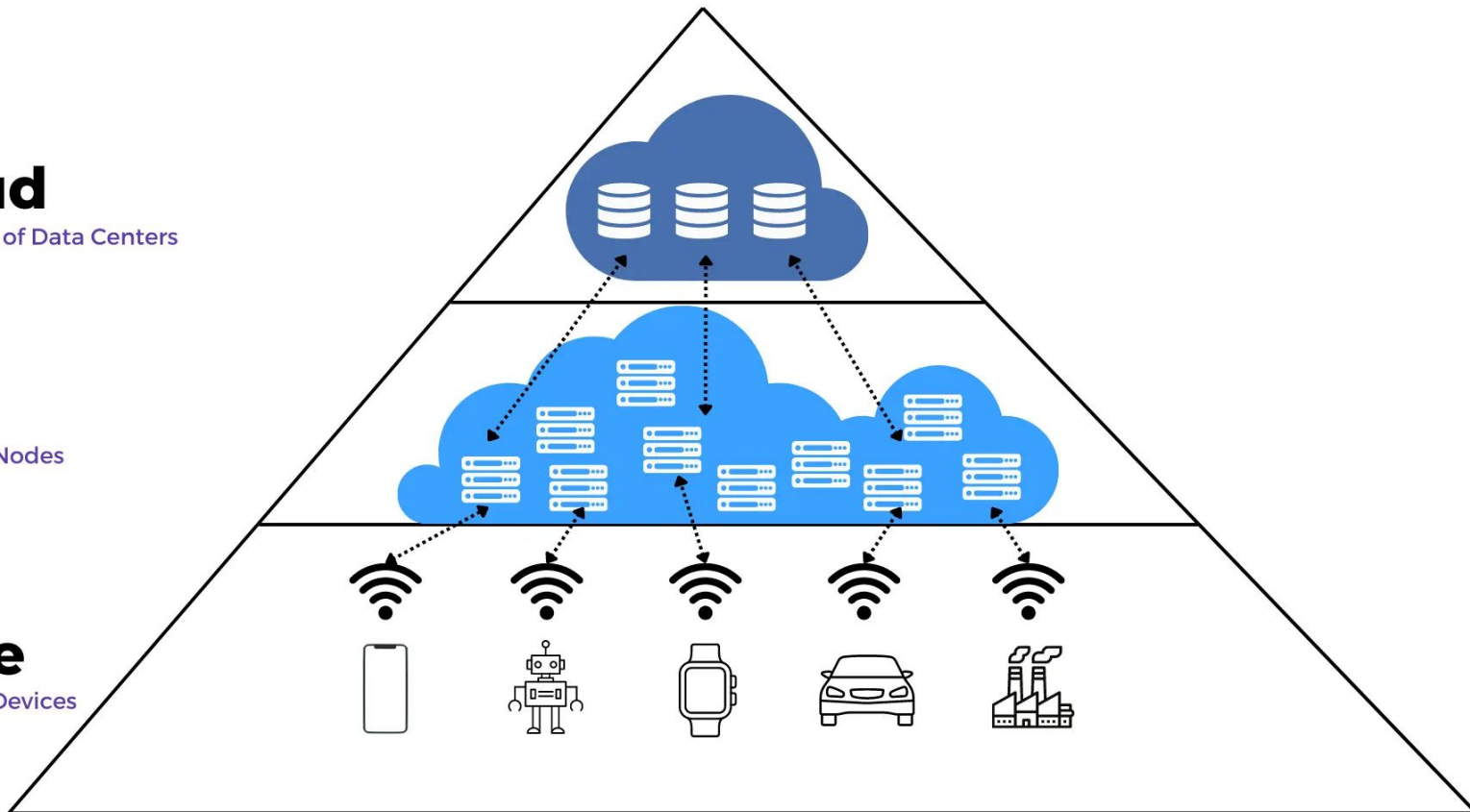
Thousands of Data Centers

Fog

Millions of Nodes

Edge

Billions of Devices



Edge layer

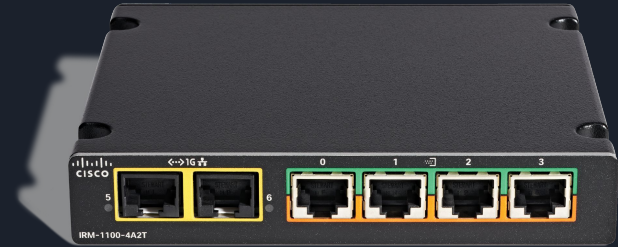
- Collects and processes data locally
- Performs real-time decisions and control
- Runs lightweight ML models on-device
- Reduces dependency on cloud and bandwidth usage

Sensors (e.g. GPS, wearables), actuators, embedded systems (such as Raspberry Pi)



Fog layer

- Aggregates data from multiple edge devices
- Performs local analytics and preprocessing
- Filters and forwards important data to cloud
- Supports real-time coordination across devices



Gateways (fog nodes),

Cloud layer

- Heavy computation and batch/stream analytics
- Long-term storage of large datasets
- Training and deployment of ML/AI models
- Device management, monitoring, and firmware updates

Cloud platforms (such as AWS), data centers, dashboards (apps)





Architectures summary

Feature	CLOUD	EDGE	FOG
Location of computing	Remote	On the device	Near the device
Latency	High	Lowest	Low
Computing power	High	Limited	Moderate
Real-time capability	Moderate	Best	Very good
Scale	Very high	Limited	Medium
Internet dependency	High	Minimal	Partial



For storing the data, choose:

- Cloud, if there is a need for long-term (historical) data

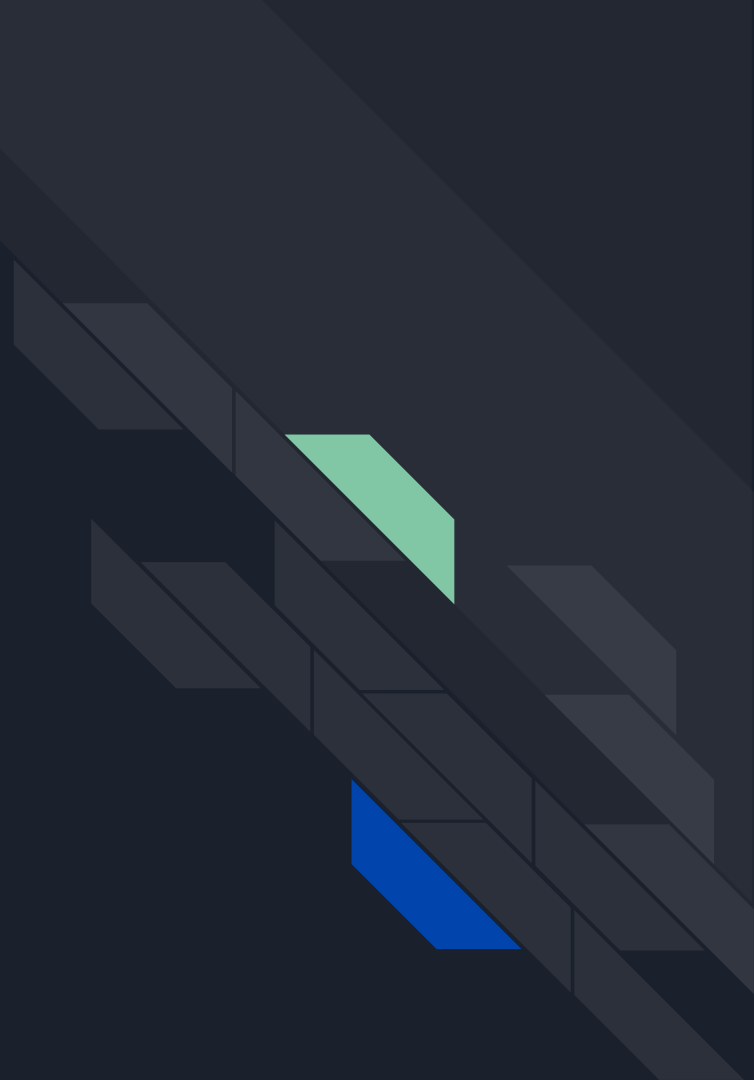


- Edge layer for real-time or short-term data

More possibilities,
but more expensive



Energy and optimisation



Battery-powered devices and low power design

Many IoT devices run on limited power.

Strategies for energy efficiency:

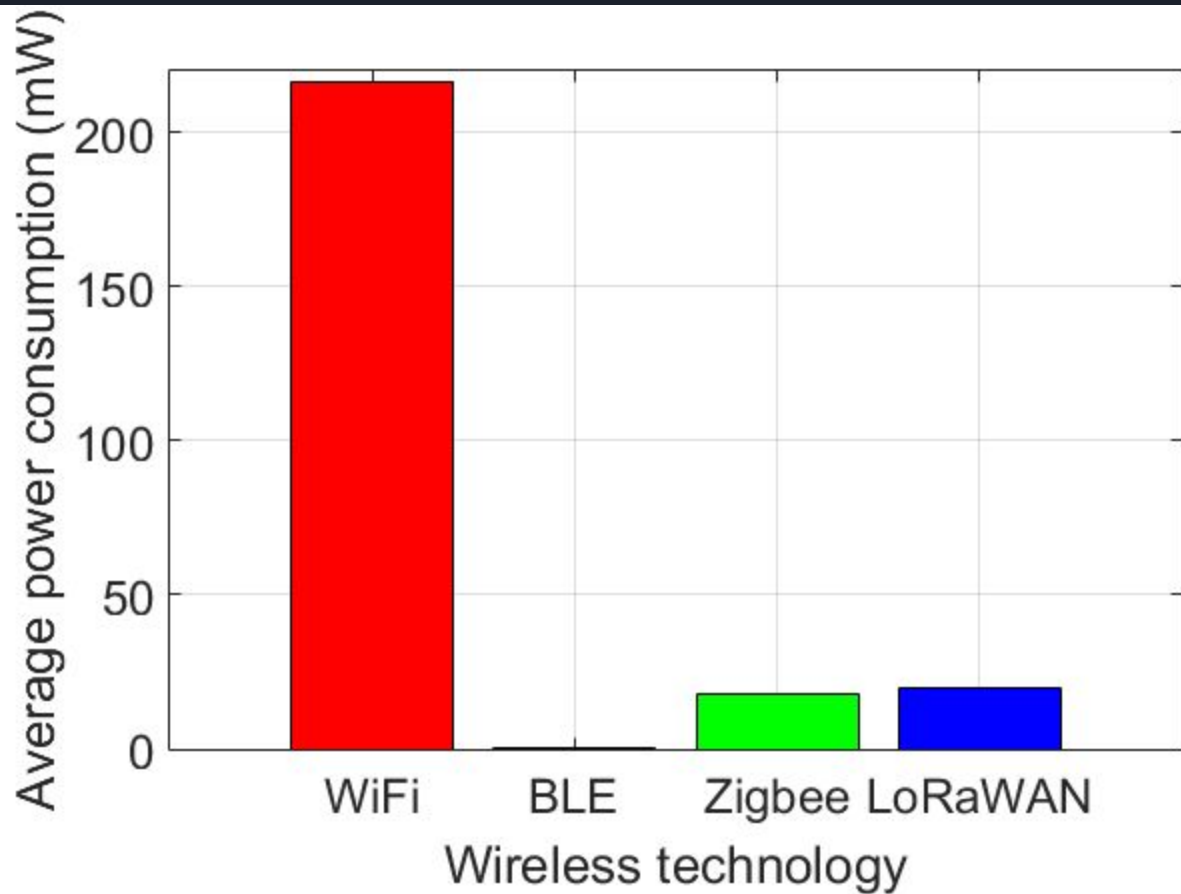
- idle state control, e.g. sleep mode
- less energy-consuming hardware
- usage of edge computing for easier tasks
- right choice of protocols



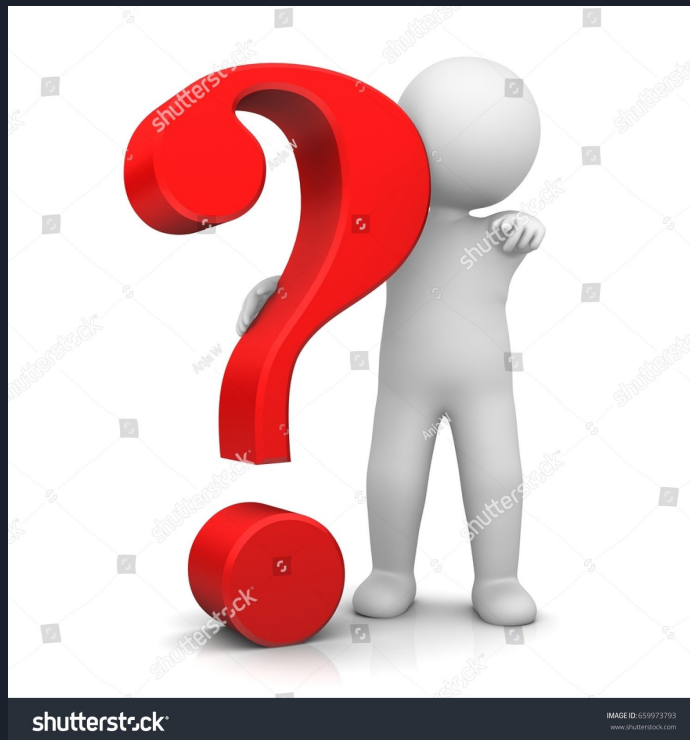


Communication protocols power comparison

Technology	Power Consumption
Wi-Fi	Medium–High
Bluetooth Low Energy (BLE)	Very Low
Zigbee	Low
LoRaWAN	Very Low
NB-IoT	Low
LTE-M	Low–Medium
5G (mMTC/URLLC)	Medium



A few questions





What protocol would you choose for:

A wearable health tracker

Technology

Wi-Fi

Bluetooth Low
Energy (BLE)

Zigbee

LoRaWAN

NB-IoT

LTE-M

5G (mMTC/URLLC)



A wearable health tracker

Technology

Wi-Fi

Bluetooth Low
Energy (BLE)

Zigbee

LoRaWAN

NB-IoT

LTE-M

5G (mMTC/URLLC)



Shipping container tracking system reaching across an entire country

Technology

Wi-Fi

Bluetooth Low Energy (BLE)

Zigbee

LoRaWAN

NB-IoT

LTE-M

5G (mMTC/URLLC)



Shipping container tracking system reaching across an entire country

Technology

Wi-Fi

Bluetooth Low Energy (BLE)

Zigbee

LoRaWAN

NB-IoT (Eventually)

LTE-M

5G (mMTC/URLLC)



Sprinkler system with a soil-moisture sensor sending data once per hour from a field 4km away?

Technology

Wi-Fi

Bluetooth Low
Energy (BLE)

Zigbee

LoRaWAN

NB-IoT

LTE-M

5G (mMTC/URLLC)



Sprinkler system with a soil-moisture sensor
sending data once per hour from a field 4km away?

Technology

Wi-Fi

Bluetooth Low
Energy (BLE)

Zigbee

LoRaWAN

NB-IoT

LTE-M

5G (mMTC/URLLC)





The other side of convenience

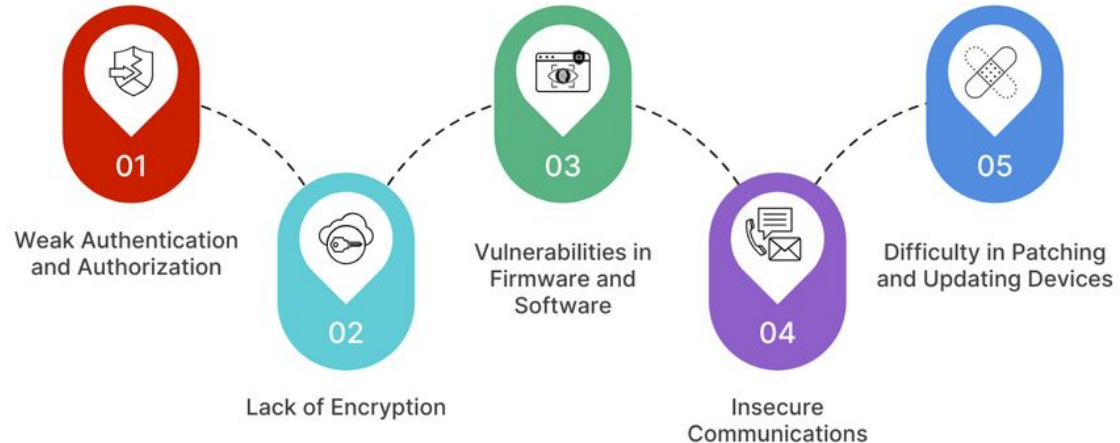
The connected world offers incredible benefits...
but it also creates a new set of challenges.

The challenges and dangers of IoT

- IoT security
- Privacy and data protection
- Legal and regulatory aspects
- Future challenges

IoT security - a massive attack surface

- Lack of standardization
- Stoppage of continuous security updates
- Weak device security
- Default passwords



IoT security - the physical threat

- 2015 Jeep hack
- Sniffing attacks
- Florida water treatment facility hack



Privacy and data protection

- Speakers
- TVs
- Robot vacuum cleaners
- Sensitive health data





Edge AI - a promising solution

Legal & Regulatory Aspects

A wooden gavel with a brass band around its middle, resting on a circular wooden block. The background is dark and out of focus, showing a hand holding a pen.

- Problems with abiding to laws:
 - right to be forgotten
 - right to data portability
- Accountability



Future challenges

- Interoperability
- Infrastructure Maintenance
- Sustainable Development and E-Waste

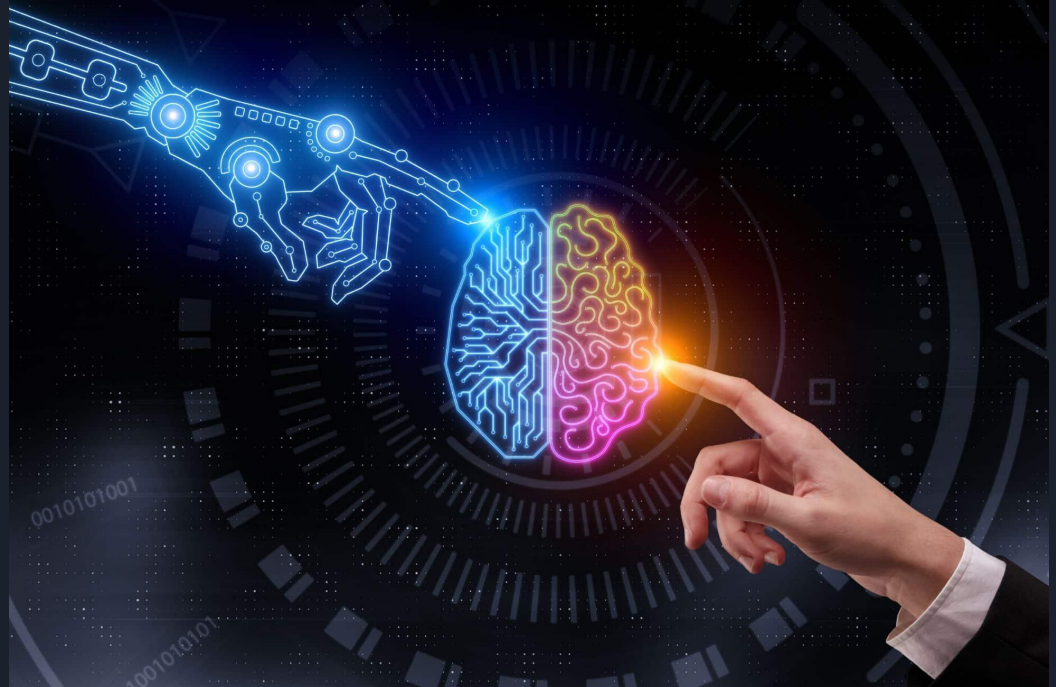


No Internet
Connection



Conclusion: Building a Responsible IoT

- Security-by-Design
- Privacy-by-Default
- Regulations





**Thank you for
listening!!!**



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